

Data Management Plans (DMP) and the Role of Libraries

IS262A

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Why DMPs Matter

Intro

- DMPs aren't paperwork. They are living guides to data reuse.
- They describe what data you'll collect, how you'll manage and preserve it, and who's responsible.
- They connect compliance, stewardship, and reproducibility.

Think of the last time you documented your data workflow. How specific were you?

The Research Data Lifecycle

Proposal → Active Research → Preservation → Reuse

- DMPs should anticipate transitions between these phases.
- Each phase has different risks and responsibilities.



Anatomy of a Strong Plan

Data Description	What will be created and in what format
Documentation	Metadata standards, README files, versioning
Storage & Backup	Where data live during the project
Sharing & access	Who can see it and under what conditions
Ethics & compliance	Privacy, consent, and IRB considerations
Roles & responsibilities	Who maintains, who deposits
Preservation	How it remains usable after the project

What makes a strong DMP

Area	Weak	Strong
Data description	“Survey data”	“De-identified survey responses, CSV, 200 records”
Storage	“Stored securely”	“Stored on UCLA Box with nightly backup”
Sharing	“Available upon request”	“Deposited in UCLA Dataverse under CC-BY license”
Documentation	“TBD”	“README with variable definitions in OSF repository”

How Librarians Support Researchers

- Reviewing and improving DMP drafts.
- Mapping repository and funder requirements.
- Providing guidance on metadata, formats, and licensing.
- Offering infrastructure: UCLA Dataverse, Redivis, Merritt.
- Teaching FAIR and open-science practices.

The Data Interview

- Like a reference interview, but focused on data.
- Ask early: What formats?
 - What volume?
 - How sensitive?
 - Who owns it?
- These questions shape both the plan and long-term access.

Common Fixes Librarians Might Make

- Clarify data type and format.
- Add preservation and backup details.
- Replace vague sharing statements.
- Align metadata with repository requirements.
- Add ethical and legal compliance details.

Finding Guidance

Need	Source	Example
Funder policies	NSF PAPPG, NIH Data Sharing policy	Identify exact requirements
Example plans	<u>DMPTool</u>	Review successful templates
Repositories	re3data.org	Match data type to repo
Standards	FAIRsharing.org	Locate metadata schemas

Common Challenges

- Vague or missing sharing statements
- Weak metadata documentation
- Unclear backup or storage strategy
- Unresolved ethical questions
- Roles and responsibilities left undefined

Case 1: Scraping Craigslist for Housing Data

The screenshot shows the Craigslist Toronto homepage. The header includes the 'craigslist' logo, a search bar, and a language dropdown set to 'english'. The main content area is divided into several columns of category links. The 'community' column lists activities, artists, childcare, classes, events, general, and groups. The 'housing' column lists apartments, housing swap, parking/storage, real estate, and vacation rentals. The 'jobs' column lists accounting/finance, admin, arch/engineering, art/media/design, biotech/science, business/mgmt, customer service, education, food/bev/hosp, general labor, government, human resources, internet engineers, legal/paralegal, manufacturing, marketing/pr/ad, medical/health, nonprofit sector, real estate, retail/wholesale, sales/biz dev, salon/spa/fitness, security, skilled trade/craft, software/qa/dba, systems/network, technical support, transport, tv/film/video, web/info design, writing/editing, and [ETC]. The 'personals' column lists strictly platonic, women seeking women, men seeking women, men seeking men, misc romance, casual encounters, missed connections, and rants and raves. The 'discussion forums' column lists a wide range of topics from apple to haiku. The 'services' column lists automotive, beauty, cell/mobile, computer, creative, cycle, event, farm+garden, financial, household, and labor/move. The 'gigs' column lists computer, event, creative, labor, crew, talent, and domestic. The 'resumes' column is also present. A sidebar on the left contains links for 'post to classifieds', 'my account', 'search craigslist', 'event calendar', and various help/faq links. A 'nearby cities' list is on the right.

community	housing	jobs
activities	apts / housing	accounting+finance
artists	housing swap	admin / office
childcare	housing wanted	arch / engineering
classes	office / commercial	art / media / design
events	parking / storage	biotech / science
general	real estate for sale	business / mgmt
groups	rooms / shared	customer service
	rooms wanted	education
	sublets / temporary	food / bev / hosp
	vacation rentals	general labor
		government
		human resources
		internet engineers
		legal / paralegal
		manufacturing
		marketing / pr / ad
		medical / health
		nonprofit sector
		real estate
		retail / wholesale
		sales / biz dev
		salon / spa / fitness
		security
		skilled trade / craft
		software / qa / dba
		systems / network
		technical support
		transport
		tv / film / video
		web / info design
		writing / editing
		[ETC]
		[part-time]

personals	for sale	services
strictly platonic	antiques	automotive
women seek women	free	beauty
women seeking men	furniture	cell/mobile
men seeking women	garage sale	computer
men seeking men	general	creative
misc romance	heavy equip	cycle
casual encounters	household	event
missed connections	jewelry	farm+garden
rants and raves	materials	financial
	motorcycles	household
	music instr	labor/move
	photo+video	
	rvs+camp	
	sporting	
	tickets	
	tools	
	toys+games	
	trailers	
	video gaming	
	wanted	

discussion forums	gigs	resumes
apple	computer	
arts	event	
atheist	creative	
autos	labor	
beauty	crew	
bikes	talent	
celebs	domestic	
comp	writing	
crafts		
diet		
divorce		
dying		
eco		
educ		
feedback		
film		
fitness		
fixit		
food		
frugal		
gaming		
garden		
haiku		

A project aimed to study rental listings and re-entry housing for formerly incarcerated tenants.

What happened:

- Craigslist's terms prohibit scraping, and the plan included collecting landlord emails—raising privacy and IRB issues.
- He halted the project until it could be redesigned with compliant methods.

Takeaway:

- Ethical and legal checks belong in every DMP review.

Case 2: Drone Imagery and Tribal Lands



A DMP proposed depositing aerial imagery in Dataverse.

The data were captured over tribal land.

Action:

We paused deposit, advised research to consult with tribal representatives, and required the CARE Principles for Indigenous Data Governance.

Takeaway: Data sovereignty and ethical deposit go hand in hand.

Case 3: Voter Data and Clinical Research



Purchased historical voter data from L2 included names and addresses. Stored in Redivis with restricted access; later combined with clinical data from UCLA Health.

Result:

Triggered new DUA and IRB reviews and required collaboration with MedNet data experts.

Takeaway: Cross-disciplinary data brings complex compliance intersections.

UCLA Workflow Example

1. Researcher drafts DMP in DMPTool.
2. Librarian reviews, adds repository, metadata, and access details.
3. Researcher reuses that DMP language for deposit or publication.

Quick Demo: DMPTool

- Log in through UCLA SSO.
- Open a funder template—NSF or NIH.
- Show how institutional fields auto-populate.
- Export a sample plan to PDF.

Question: Which section do you think most researchers struggle to complete?

Connecting DMPs to Repositories

- DMPs are roadmaps that lead to actual deposits.
- UCLA repositories enable FAIR principles:
 - **Dataverse:** open, citable data. At UCLA:
<https://dataverse.ucla.edu/>
 - Dryad: <https://datadryad.org/>
 - **Redivis:** *restricted or sensitive data*:
<https://redivis.com/UCLALibrary>
 - **Merritt:** preservation storage: <https://merritt.cdlib.org/>

Metadata and Documentation

- Metadata is what turns data into knowledge.
- Key elements librarians ensure:
 - Persistent identifiers (DOI)
 - Controlled vocabularies
 - Reproducible documentation

Example: Dataverse “Terms” tab → aligns with DMP fields.

Repository Selection Exercise

Ask: Which repository fits these datasets?

1. Sensitive clinical interviews
2. 3D drone imagery
3. Open survey CSVs

Use re3data.org to find matches

Also, NIH Data Sharing Resources, Domain Specific and Generalist

https://www.nlm.nih.gov/NIHbmic/domain_specific_repositories.html

Repository comparison: <https://zenodo.org/records/3946720>

Credits and Contact

Adapted from Library Carpentry DMP101 (CC BY 4.0)

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